MOAIN AL-ABBASI

Full-Stack Software Engineer · AI Engineer (LLM Agents, LangGraph, MCP)

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SUMMARY

Software engineer and founder of Alabbasi Tech, building business systems and AI agents end to end in TypeScript, Python and Go. Built a POS on an existing Oracle database with Bluetooth receipt printing, a ZKTeco attendance system with Saudi WPS payroll export, a multi-tenant notification platform, LangGraph multi-agent systems and MCP servers, and Basira, an Arabic quotation verifier with 0/500 false alarms on its evaluation set. Fully remote, Saudi time zone.

EXPERIENCE

Founder & Software Engineer · Alabbasi Tech (alabbasi.uk) · Sana'a, Yemen · Remote — 2026 – Present

- **POS replacing YemenSoft Onyx Pro** — built a web POS on the existing Oracle database (no migration): sales, returns, shifts and reports. Arabic receipts print on a Bixolon SPP-R310 over Bluetooth SPP through Web Serial; documented why Xprinter USB fails on Windows (driver). — NestJS 10, React 19, Oracle PL/SQL, OpenAPI
- **Motech Attendance** — built to replace the ZKTeco desktop software: branch collectors pull punches from iFace950 devices over TCP into a central multi-tenant service with idempotent ingest; shifts, leave balances, CSV/Excel reports and WPS payroll export for Saudi Arabia (MHRSD/SAMA) and the UAE (SIF). Tested on a live device (3,118 punches). — Go, PostgreSQL, Python (pyzk), systemd
- **Nexus Notify** — designed a multi-tenant, multi-channel notification platform behind one API, with an Angular dashboard and a native Android SMS gateway (dual SIM, foreground service). — NestJS, PostgreSQL 17, Prisma, Redis/BullMQ, Angular 22, Kotlin, Jetpack Compose
- **Secure remote access** — built remote access to managed machines: Authenticode-signed Windows agent running as a service, NetBird mesh, SSH key rotation, private keys encrypted with AES-256-GCM and a full activity log; cross-platform CLI with cosign-signed releases from GitHub Actions. — Go, PostgreSQL, SSH, NetBird
- **Google Ads connector for Keif Al-Diafa (Jeddah)** — built a permanent OAuth connection to Google Ads and Search Console with free-form GAQL queries and an Arabic dashboard (phase 0 complete). — Hono, Cloudflare D1
- **Multi-tenant ads platform** — completed phase 1 of a SaaS for managing Google Ads across several clients: live API data, a 12-table schema and a dashboard, with human approval required before any change. — Next.js 15, Prisma, PostgreSQL 16
- **keif-publisher** — built an engine that generates posts from a media archive, routes each through a human approval gate, then publishes on schedule; dry-run and kill switch on by default. — Python, FastAPI, APScheduler
- **Saudi hospitality brands** — websites keifaldiafa.com and asoulaldiafa.com (Next.js 14); Android apps for Keif Al-Diafa and Asoul Al-Diafa (v2.6.0, signed releases with remote lock). — Next.js 14, Flutter, Kotlin

AI AGENTS & APPLIED AI

Basira · basirapp.site · github.com/MoTechSys/Project-Basira

Verifies Quran and Hadith quotations byte-exactly against 71,987 records from licensed corpora; an optional LLM only proposes spans and reads images (OCR), and never decides a verdict. 150/150 evaluation cases, 0/500 false alarms, 78.54% on IslamicEval 2025 subtask 1B (public dev set). 487 automated tests (345 backend, 110 bot, 32 frontend). REST API, MCP server and Telegram bot; entry to the AI in Service of Islamic Content Challenge 2026. — Python, FastAPI, React 19

Mansur AI (private repository)
Arabic multi-agent system: planner, ReAct executor, verifier and learner orchestrated with LangGraph; 13 agents, 38 tools, model router with fallback and cost tracking, hierarchical memory (working, episodic, procedural, semantic) and skills saved from successful runs. 82 passing tests. — Python, LangGraph, FastAPI, WebSocket, SQLite

Ai_Alabbasi (private repository)
Coding agent with a model backend swapped from one config file (Claude, GPT-5, DeepSeek or a local model); task planner, CodeAct sandboxed Python, hybrid retrieval with FTS5 and vectors fused by RRF. — Python, SQLite FTS5

Moain Promoter AI
Seven specialised agents under an orchestrator write bilingual Arabic/English posts for six platforms, with swappable model providers and a Next.js dashboard. — Python, Celery, Redis, Playwright

SELECTED PROJECTS

Idhaat · github.com/MoTechSys/Idhaat-Platform
Live-tutoring platform: allocates four Zoom rooms with fallback to self-hosted Jitsi (JWT per lesson); recordings upload to Cloudflare R2 during class, carry a moving watermark with the student's name and are deleted after 48 hours. Runs for about 25 SAR a month. — Hono, Cloudflare Pages, D1, R2, Vitest

Grocery Ledger · github.com/MoTechSys/mohanad-web-app-2
Offline Android accounting and POS app for small shops: barcode by camera (ML Kit) or USB/Bluetooth scanner, append-only ledgers, PDF/Excel export. 157 passing tests, v2.2.1. — Flutter

Sijil · github.com/MoTechSys/Sijilati
Offline debt ledger for shop owners and workers, including those who cannot read: amounts entered as banknotes, spoken confirmation, 8-second undo. 93 passing tests. — Flutter

Ghaida · github.com/MoTechSys/ghaida
Household guidebook for domestic workers in their native language with AI voice for every sentence; offline PWA with admin panel.

S-ACM · github.com/MoTechSys/S-ACM-Project
Academic content-management system with AI summarisation and question generation. — React 19, TypeScript, Tailwind 4

GRADUATION PROJECT (TEAM)

Security Operations Center (SOC) · github.com/MoTechSys/my-bro
SOC built on open-source Wazuh 4.14.7, Suricata, YARA and auditd, integrated with VirusTotal and MikroTik syslog, with an AI analyst whose actions require human approval. 14 use cases verified in a lab, including automatic blocking of SSH brute-force attacks. Digital forensics lab with Autopsy and FTK Imager.

TECHNICAL SKILLS

Programming	TypeScript, Python, Go, Dart, Kotlin, SQL, PL/SQL
Backend	NestJS, FastAPI, Hono, Prisma, REST/OpenAPI, WebSockets, Redis, BullMQ, Celery
Frontend & mobile	React 19, Next.js, Angular 22, Tailwind CSS, PWA, IndexedDB, Arabic RTL; Flutter, Jetpack Compose, ML Kit
Data	PostgreSQL, Oracle, SQLite, Cloudflare D1
AI & agents	LangGraph, multi-agent orchestration, MCP servers, RAG, hybrid search (FTS5, vectors, RRF), model routing; OpenAI, Anthropic, Gemini
Integration	ZKTeco (pyzk), Arabic raster printing over Bluetooth SPP and Web Serial, Google Ads API, Search Console
Security	OAuth, JWT, AES-256-GCM, SSRF protection, sandboxed execution, Authenticode and cosign signing; Wazuh (SIEM), Suricata (IDS), YARA, Autopsy
DevOps	Docker, GitHub Actions, Cloudflare (Pages, D1, R2), Linux; pytest, Vitest, Playwright
Languages	Arabic (native), English (technical reading and writing)